

Extension Activity: Exploring Data Types with User Input

Instructions: Design a short Python program named **03_extension_task** that asks the user a few questions, does something with the answers, and prints a clear summary. This is your chance to explore something new and use your knowledge of data types.

Step 1: Research and Explore

A) How to use `input()`

Structure:

```
variable_name = input("Write a prompt for the user: ")
```

Whatever the user types is stored as a **string**.

B) Data type conversion with `int()` and `float()`

Because `input()` gives you a string, convert when you need numbers.

```
age = int(input("How old are you? "))
price = float(input("Enter a price: "))
```

C) Lowercasing and building a boolean from text

Turn user text into lowercase, then compare it to make a True/False value.

```
is_student_text = input("Are you a student? (yes/no): ").lower()
is_student = (is_student_text == "yes")
print("Student status:", is_student)
```

D) Printing clearly with commas or concatenation

Commas automatically add spaces and do not need conversion:

```
print("Age in years:", age)
```

Concatenation needs strings, so convert numbers first:

```
print("Age in years: " + str(age))
```

E) Checking types with `type()`

```
print(type(age))      # <class 'int'>
print(type(price))    # <class 'float'>
print(type(is_student)) # <class 'bool'>
```

Step 2: Create Your Program

Program Requirements

1. **Collect user input**
 - Ask for at least **three** pieces of information.
 - Include at least **one string**, **one integer**, and **one float**.
2. **Do a calculation or operation**
 - Use one or more numeric inputs to compute something useful or fun.
 - Examples: total cost, value doubled, hours to minutes.
3. **Display a summary**
 - Print a short, user-friendly summary that includes the original inputs and your result.
 - Use commas or concatenation.
 - Show at least one call to `type()` so you can prove a conversion worked.

Example Scenario: Personal Fun Facts Generator

What it collects

- Name (string)
- Age (integer)
- Favorite number (float)

What it calculates

- Age in 10 years
- Favorite number doubled

Example interaction

```
What is your name? Alex
How old are you? 16
What is your favorite number? 3.14
```

Fun Facts About You:

```
Name: Alex
Age in 10 years: 26
Your favorite number doubled: 6.28
```

Starter template (fill in and modify):

```
# Collect inputs
name = input("What is your name? ")
age = int(input("How old are you? "))
fav_num = float(input("What is your favorite number? "))

# Show types for verification
print("Type of name:", type(name))
print("Type of age:", type(age))
print("Type of fav_num:", type(fav_num))

# Do calculations
age_in_10 = age + 10
fav_num_doubled = fav_num * 2

# Summary (use commas or concatenation)
print()
print("Fun Facts About You:")
print("Name:", name)
print("Age in 10 years:", age_in_10)
print("Your favorite number doubled:", fav_num_doubled)

# Optional: build a boolean from text
student_text = input("Are you a student? (yes/no): ").lower()
is_student = (student_text == "yes")
print("Student status:", is_student)
print("Type of is_student:", type(is_student))
```

Try One of These Themes Instead

- **Snack Stand Helper:** item name, number of items, price per item, calculate total.
- **Study Timer:** subject, study minutes, convert to hours, report a summary.
- **Workout Log:** activity, minutes, calories per minute, total calories.
- **Game Stats:** gamer tag, matches played, average score, total score estimate.
- **SMART Goal Tracker:** goal, times you completed the task this week, points earned per task, total points for the week

Submission Checklist

- I collected at least three inputs, including one string, one int, and one float.
- I converted user input to the correct types with `int()` or `float()`.
- I used `type()` at least once to verify a conversion.
- I performed one calculation with numeric inputs.
- I printed a clear summary using commas or concatenation.
- My code runs without errors.

Reflection Questions

1. What was the trickiest part of using `input()` with numbers, and how did you fix it?
2. How did you decide which data type to use for each input?
3. If you had more time, what feature would you add next?

Optional Extensions

Continue to do some research on your own (using Google) to see the way to implement some of the following into your code:

- Add simple validation, such as asking again if the user types a non-number for age.
- Add a boolean derived from text using `.lower()` and a comparison to "yes" or "no".
- Preview idea only: write a comment that shows how your summary could look later with an f-string.